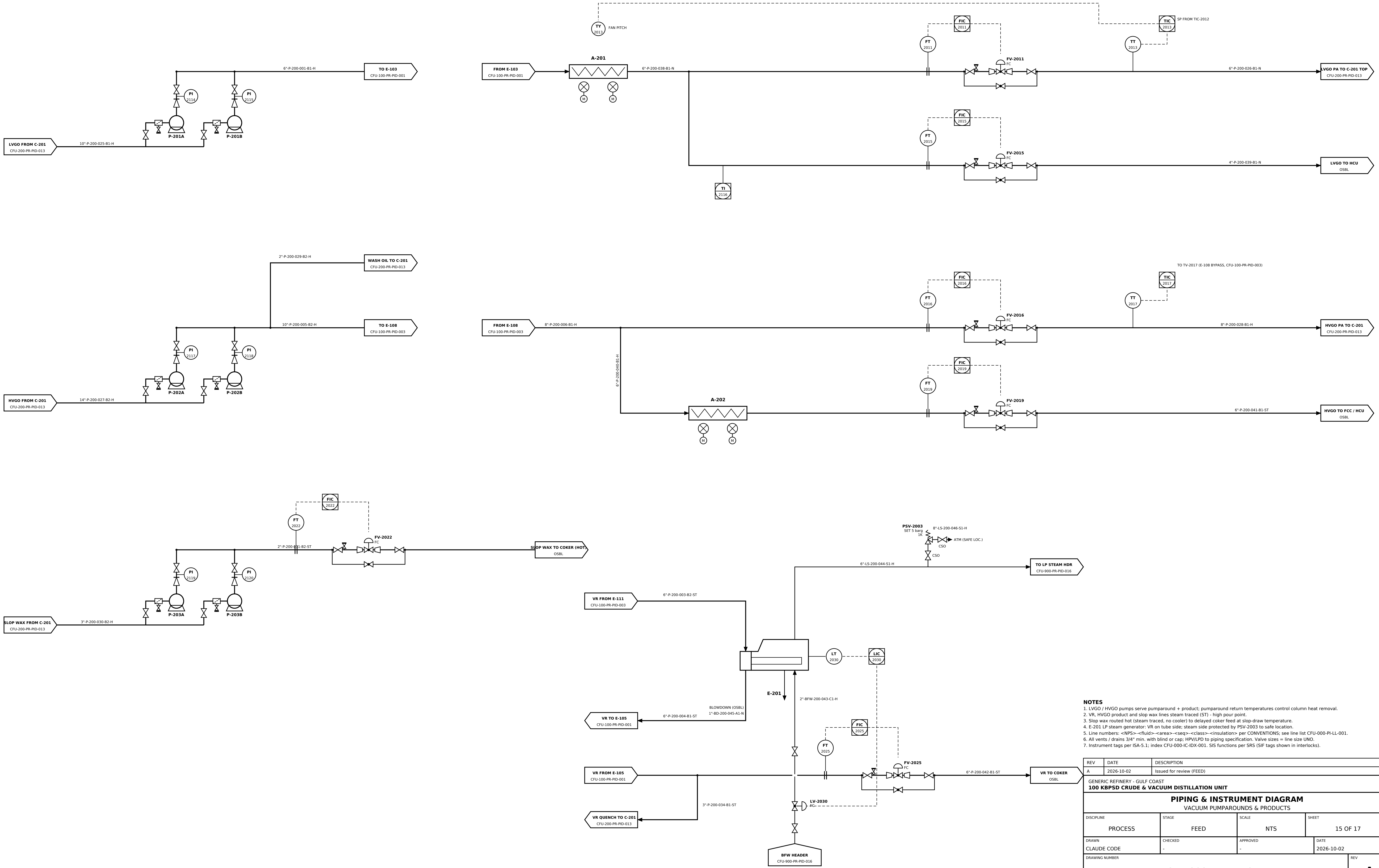


P-201A/B LVGO pumparound / product RATED 198 m3/h @ 138 m, dP 10 bar MOTOR 110 kW, API 682 Plan 1122 DES. T 255 C; MAT'L S-6 (CS/12Cr)	P-202A/B HVGO pumparound / product RATED 391 m3/h @ 144 m, dP 10 bar MOTOR 200 kW, API 682 Plan 538 DES. T 345 C; MAT'L A-8 (316)	P-203A/B Slop wax RATED 12 m3/h @ 114 m, dP 8 bar MOTOR 5.5 kW, API 682 Plan 538 DES. T 410 C; MAT'L A-8 (316)	A-201 Lvgo cooler 397 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 9.7 barg / 125 C DUTY 3.10 MW; 2 x 15 kW fans MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)	A-202 Hvgo Product cooler 361 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans DESIGN 9.7 barg / 125 C DUTY 6.64 MW; 2 x 37 kW fans SHELL: BFW / LP steam / TUBE: Vacuum residue MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional)	E-201 VR / LP steam generator (kettle) 84 m2; 1 shell(s) DESIGN 6 / 20 barg / 310 C; CA 3 mm DUTY 1.78 MW SHELL: BFW / LP steam / TUBE: Vacuum residue MAT'L CS shell / Scr tubes
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1. LVGO / HVGO pumps serve pumparound + product; pumparound return temperatures control column heat removal.

2. VR, HVGO product and slop wax lines steam traced (ST) - high pour point.

3. Slop wax routed hot (steam traced, no cooler) to delayed coker feed at slop-draw temperature.

4. E-201 LP steam generator: VR on tube side; steam side protected by PSV-2003 to safe location.

5. Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.

6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.

7. Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING & INSTRUMENT DIAGRAM
VACUUM PUMPAROUNDS & PRODUCTS

DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	15 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER	REV		
CFU-200-PR-PID-014			
A			